Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_15

1. Summary of Key Deviations

The analysis of Run_15 reveals a significant physical fault characterized by a major over-delivery of fuel, which the system is struggling to manage.

- **Flow_Bypass (Moderate):** The bypass flow shows a large positive **offset** of approximately **+25%**.
- **Flow_Pump (Moderate):** The pump flow also shows a significant positive **offset**, reading **+14.5%** higher than the healthy baseline. This indicates the pump is delivering substantially more fuel than commanded.
- **p_Combustion (Moderate):** The combustion pressure shows a major **negative offset**, reading around **-18%** lower than healthy. This is a critical and counter-intuitive symptom.
- **Flow_PRV (Moderate):** Uniquely, the `Flow_PRV` signal shows a moderate positive **offset** of **+13.8%** at the very beginning of the run (state s1), indicating the high-pressure relief valve is active even at idle.
- **Other Signals (Nominal/Minor):** All other signals are either nominal or show only minor, likely secondary, deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump & Combustion Chamber**	**Dual Fault: Pump Over-performance & Combustion Leak**	This is the only hypothesis that explains the contradictory evidence. There are two independent faults: 1) The **Main Fuel Pump is significantly over-performing** (+14.5%), causing the high `Flow_Pump` and `Flow_Bypass` readings. 2) A **major leak exists in the Combustion Chamber** or at an injector nozzle, causing the large `p_Combustion` pressure drop (-18%) despite the high fuel flow. The PRV activation at idle further supports a system struggling with excess flow/pressure from the start.

2	**Main Fuel Pump**	**Significant Over-performance**	This explains the high `Flow_Pump` and `Flow_Bypass` readings. However, it cannot explain why `p_Combustion` is severely *low*. High flow should lead to high pressure, not low pressure.
3	**Combustion Chamber / Injector Nozzle(s)**	**Major Leak**	This explains the major drop in `p_Combustion`. However, a leak alone would not cause the `Flow_Pump` to read +14.5% higher than normal.
4	**p_Combustion Sensor**	**Sensor Failure (Negative Offset)**	A failed sensor could explain the low `p_Combustion` reading. This would make the high `Flow_Pump` reading a sign of pump over-performance (a real fault). This is a plausible dual-fault scenario, but a dual *physical* fault (Hypothesis 1) is often more likely in such complex systems.
5	**Pressure Relief Valve (PRV)**	**Stuck Partially Open / Low Cracking Pressure**	The PRV is active at idle, which is abnormal. If it's stuck partially open, it would create a leak path, explaining some pressure loss. However, the `Flow_PRV` deviation is not large enough to explain the massive -18% drop in combustion pressure, and it doesn't explain the pump's over-performance.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Significant Over-performance of the Main Fuel Pump AND (2) a separate, Major Leak in the Combustion Chamber or at an Injector Nozzle.**

Reasoning: No single fault can explain the contradictory data. The system is clearly suffering from two simultaneous problems. The first is fuel over-delivery, evidenced by the high `Flow_Pump` (+14.5%) and `Flow_Bypass` (+25%) readings. The second is a major loss of pressure where it matters most, in the combustion chamber (`p_Combustion` is -18%). A system cannot have both high flow and low pressure without two things going wrong: something is making too much fuel, and something else is letting it escape before it can build pressure. The PRV activation at idle is a key clue, suggesting the system is fighting an over-supply issue from the very start.